DATA SCIENCE

Beginner-to-Intermediate Course Guide

A practical roadmap with concepts, Python examples, projects, practice tasks and video resources.

For: Students and beginners
Suggested pace: 8–12 weeks

How to use this guide

Follow the modules in order. Read the concepts, practise the code, complete the tasks and build projects. Learning by doing is essential.

Recommended study routine

- Study one concept for 30–45 minutes.
- Run examples yourself and change the values.
- Spend at least 30 minutes solving exercises.
- Keep notes of errors, definitions and useful code.
- Explain each topic in your own words.

Course roadmap

Module	Topics	Time
1. Foundations	Data science workflow and problem framing	3–4 days
2. Python	Variables, conditions, loops, functions, collections	1–2 weeks
3. Data libraries	NumPy, Pandas, importing and inspecting data	1 week
4. Statistics	Descriptive statistics, probability, distributions	1 week
5. Data preparation	Missing values, duplicates, outliers	4–5 days
6. Visualization	Matplotlib, Seaborn, charts and storytelling	1 week
7. Machine learning	Regression, classification, clustering, evaluation	2 weeks
8. Projects	End-to-end analysis and presentation	1–2 weeks

1. Introduction to Data Science

What is data science?

Data science combines statistics, programming, domain knowledge and communication to turn data into useful insights. A typical project starts with a question, gathers relevant data, prepares it, analyses it, builds a model when useful and communicates the result.

Typical workflow

Define the problem → collect data → understand the data → clean and prepare it → explore and visualise → model if needed → evaluate → communicate and monitor.

Common roles

A data analyst focuses on querying, reporting and insights. A data scientist often combines analysis with statistical or machine-learning modelling. A data engineer builds and maintains data pipelines.

Practice task

Choose a question such as: Which product category has the highest sales? List the fields you need and explain how the answer could help a business.

2. Python Fundamentals

Core concepts

Learn variables and types (int, float, str, bool), operators, input/output, if/elif/else, loops, functions, lists, tuples, sets, dictionaries, file handling and exceptions.

Example: average score

```
scores = [78, 85, 92, 67]
average = sum(scores) / len(scores)
print(round(average, 2))
```

Output: 80.5

Example: count items

```
items = ["pen", "book", "pen"]
counts = {}
for item in items:
  counts[item] = counts.get(item, 0) + 1
print(counts)
```

Output: pen: 2, book: 1

Practice tasks

Find the largest number in a list; count vowels in a sentence; calculate monthly expenses; read a CSV file and print its first five rows.

3. NumPy and Pandas

NumPy

NumPy provides efficient arrays and numerical operations. Learn array creation, indexing, slicing, shape, data types, aggregation and vectorised calculations.

Pandas

Pandas works with tabular data. Learn Series, DataFrame, read_csv(), head(), info(), describe(), loc, iloc, filtering, sorting, groupby(), merge() and exporting CSV.

Example: group sales

```
import pandas as pd
df = pd.DataFrame({"city": ["Chennai", "Madurai", "Chennai"], "sales": [120, 90, 150]})
print(df.groupby("city")["sales"].sum())
```

Result: Chennai 270; Madurai 90.

Practice task

Load a sales CSV file. Display dimensions, column names and summary statistics. Find the top five rows by sales and calculate total sales by city.

4. Statistics and Probability

Descriptive statistics

Mean is the arithmetic average. Median is the middle value after sorting. Mode is the most frequent value. Range is maximum minus minimum. Variance and standard deviation measure spread.

Probability

Probability measures how likely an event is, from 0 (impossible) to 1 (certain). Learn sample spaces, independent events, conditional probability and common distributions.

Correlation

Correlation describes the strength and direction of association between two variables. Correlation does not prove that one variable causes the other.

Practice task

For exam scores, calculate mean, median, mode, range and standard deviation. Explain which measure best represents a typical score.

5. Data Cleaning and Preparation

Why it matters

Real-world data may contain missing values, duplicates, inconsistent labels, incorrect data types and extreme values. Poor-quality input can lead to misleading conclusions.

Key techniques

Inspect missing values; remove or impute them thoughtfully; identify duplicates; standardise text; convert dates and numeric columns; investigate outliers; validate ranges.

Cleaning checklist

Check row and column counts, data types, missing values, duplicate rows, minimum and maximum values, and inconsistent categories. Document every change.

Practice task

Create a messy dataset with missing ages, repeated rows and inconsistent city names. Clean it and write down each decision.

6. Data Visualization

Choosing a chart

Use a bar chart to compare categories, a line chart for trends over time, a histogram for distributions, a scatter plot for relationships and a box plot to compare spread.

Tools

Matplotlib provides flexible plotting. Seaborn offers statistical visualisations. Power BI and Tableau are popular tools for interactive dashboards.

Good practice

Use a clear title, label axes, include units, choose readable scales, avoid unnecessary decoration and state the main takeaway.

Practice task

Create a line chart of monthly sales, a bar chart of sales by category and a histogram of order values. Explain each chart.

7. Introduction to Machine Learning

What is machine learning?

Machine learning trains a model to find patterns and make predictions or group observations. It is one part of data science, not the whole field.

Main types

Supervised learning uses labelled examples. Regression predicts numeric values; classification predicts categories. Unsupervised learning finds structure in unlabelled data, such as clusters.

Basic workflow

Define target and features → split data into training and test sets → fit a baseline model → evaluate on unseen data → improve carefully → communicate limitations.

Common algorithms

Linear regression, logistic regression, decision trees, random forests, k-nearest neighbours and k-means clustering.

Evaluation

For regression, learn MAE, MSE and RMSE. For classification, learn accuracy, precision, recall, F1-score and the confusion matrix.

Important caution

Avoid data leakage: test-set information must not influence training or preprocessing. A high score alone does not prove a model is useful or fair.

8. Portfolio Projects

Project A: Student performance

Explore how study time, attendance and other available factors relate to student scores. Use descriptive statistics and visualisations. Avoid claiming causation from simple correlations.

Project B: Retail sales analysis

Clean transaction data, calculate revenue by month and category, identify top products and build a dashboard with business recommendations.

Project C: House price prediction

Prepare property features, establish a baseline, train a regression model and evaluate predictions using MAE or RMSE. Explain limitations.

Project D: Customer segmentation

Prepare customer-level features, scale suitable numeric variables and experiment with k-means. Describe clusters carefully.

Portfolio checklist

Include the problem statement, dataset source and licence, cleaning steps, analysis, methods, evaluation, limitations, visuals and reproducibility instructions.

9. Tools and Learning Resources

Python and notebooks

Python official website · Jupyter · Google Colab

Data libraries

NumPy learning · Pandas getting started · Matplotlib tutorials · Seaborn tutorial

Machine learning

Scikit-learn getting started · Google Machine Learning Crash Course

Practice datasets

Kaggle datasets · UCI Machine Learning Repository · India Open Government Data

Video learning links

Data Science full courses
Python for Data Science
NumPy and Pandas
Machine Learning

These video links open search results so you can choose an instructor, language and course length. Check the licence before downloading or re-uploading any video. Linking to a video is different from copying it to your platform.

10. Final Revision Checklist

- I can explain the data science workflow and distinguish analysis from machine learning.
- I can write basic Python programs using functions and collections.
- I can load, inspect, filter, group and combine data with Pandas.
- I understand mean, median, standard deviation, probability and correlation.
- I can identify and handle missing values and duplicates.
- I can choose and create suitable charts.
- I can train a simple model and evaluate it on unseen data.
- I have completed at least two projects and documented their limitations.

Suggested next step

Start with Python and Pandas, then analyse a small CSV dataset. Build one clear project before moving to more complex machine-learning models.

Educational material. Adapt the pace and project scope to your learners. Check dataset permissions, privacy requirements and video licences before publishing.